

HEALTHCARE DATA CLIENT

Data Quality Rule Catalogue

Validation Rule Governance and Exception Classification

Document Control	Detail
Engagement	Jul-Oct 2026
Role	Data Analyst
Primary Stakeholder	Healthcare Analyst
Processing Scale	Approximately 250,000 records per batch
Confirmed Validation Areas	Duplicates, nulls, dates, formats, IDs, ranges, codes and cross-field mismatches
Observed Exception Rate	2%
Rule catalogue status: The validation categories are confirmed. Exact historical field names, thresholds, code sets and rule expressions are not retained, so this catalogue defines governed rule patterns without inventing client logic.	

1. Rule Catalogue

Rule ID	Rule	Dimension	Purpose	Applies To	Rule Parameter	Reason Category
DQ-01	Duplicate Detection	Uniqueness	Identify records meeting the documented duplicate definition.	Record or defined field combination	Duplicate definition must be specified before execution.	Duplicate / uniqueness failure
DQ-02	Required Value Check	Completeness	Identify null, blank or missing values in fields subject to completeness rules.	Applicable required field	Required-field scope must be documented.	Missing required value
DQ-03	Date Validity Check	Validity	Identify date values that cannot be validly interpreted under the documented date rule.	Applicable date field	Parsing rule and permitted representation must be documented.	Invalid date
DQ-04	Format Check	Conformance	Identify values that do not match a documented expected format.	Applicable formatted field	Expected format must be documented.	Incorrect format
DQ-05	Identifier Validity Check	Validity	Identify identifiers failing the documented identifier rule.	Applicable identifier field	Identifier rule must be documented. No healthcare ID pattern is assumed.	Invalid identifier
DQ-06	Range Check	Validity	Identify numeric or date values outside documented valid bounds.	Applicable numeric or date field	Lower/upper bounds must be documented before execution.	Out-of-range value
DQ-07	Code Consistency Check	Consistency	Identify values inconsistent with a documented valid code set or mapping.	Applicable code/category field	Valid code set or mapping must be documented.	Inconsistent code
DQ-08	Cross-Field Consistency Check	Integrity	Identify records where documented relationships between two or more fields are not satisfied.	Related field set	Cross-field relationship must be explicitly defined.	Cross-field mismatch

2. Exception Classification Rule

Element	Governed Definition
Exception Record	A processed record failing one or more applicable validation rules.
Exception Flag	Record-level indicator that at least one applicable rule failed.
Reason Code	Code or label identifying the validation failure.
Validation Failure	A rule-level failure associated with a processed record.
Exception Output	Separate file, table or report containing records requiring review.
Exception Rate	Exception Records / Records Processed x 100.

Observed result: The engagement surfaced a 2% exception rate. This is not a validation threshold, acceptable-error limit or SLA.

3. Rule Execution Principles

Principle	Control
Applicability	Run a rule only against fields for which the rule is defined.
Parameter governance	Do not infer thresholds, code sets, formats or cross-field relationships.
Failure traceability	Capture the affected source record and the rule/reason associated with the failure.
Multiple failures	Allow one record to carry multiple rule-level failures.
Record counting	Count an affected record once in Exception Records regardless of the number of failures.
No silent repair	Do not fabricate or overwrite questionable healthcare values solely to make validation pass.
Repeatability	Apply the same documented rule logic consistently across recurring batches.

4. Rule Specification Template

Specification Element	Required Content
Rule ID	Unique rule reference
Business Name	Reviewer-readable rule name
Quality Dimension	Uniqueness, completeness, validity, conformance, consistency or integrity
Source Field(s)	Actual source field or field combination
Rule Logic	Documented pass/fail condition
Parameters	Expected format, valid range, code set or relationship where applicable
Exception Reason	Reason code or readable failure explanation
Output Grain	Record-level or rule-level
Test Evidence	Positive, negative and boundary test cases where applicable
Owner / Reviewer	Responsible business or data reviewer where formally defined

5. Rule-Specific Test Conditions

Rule	Minimum Test Condition	Expected Result
DQ-01 Duplicate	Known duplicate and known non-duplicate examples under the implemented definition	Only records satisfying the documented duplicate rule are flagged.
DQ-02 Missing Value	Null/blank and populated values	Only applicable missing values fail.
DQ-03 Date	Valid and invalid date representations under the implemented rule	Invalid values are flagged without inventing replacement dates.
DQ-04 Format	Matching and non-matching values	Non-conforming values are flagged.
DQ-05 Identifier	Valid and invalid examples under the documented identifier rule	Only values failing the defined rule are flagged.
DQ-06 Range	Inside-range and outside-range values, including boundaries	Values outside documented bounds are flagged.
DQ-07 Code	Allowed and non-allowed values under the documented code set	Values outside the governed set or mapping are flagged.
DQ-08 Cross-Field	Records satisfying and violating the documented relationship	Only relationship violations are flagged.

6. Reason Code Governance

Control	Requirement
Uniqueness	Each implemented validation rule should map to an identifiable reason or failure description.
Clarity	Reason wording should explain why the record requires review.
Consistency	The same rule should use consistent reason classification across batches.
Multiplicity	A record may have multiple reasons when multiple rules fail.
Reporting	Reason categories may support exception analysis when the underlying output preserves rule-level detail.
Historical boundary	Exact client reason-code vocabulary is not retained and must not be invented as historical evidence.

7. Rule-to-BRD Traceability

Rule	BRD Requirement	User Story
DQ-01	BR-03	US-02
DQ-02	BR-04	US-02
DQ-03	BR-05	US-03
DQ-04	BR-05	US-03
DQ-05	BR-06	US-03
DQ-06	BR-07	US-03
DQ-07	BR-08	US-04
DQ-08	BR-09	US-04
Exception classification	BR-10 to BR-13	US-05, US-06
Exception Rate	BR-14	US-07

8. Rule Output & Reconciliation

Output	Grain	Purpose
Processed Record Output	One row per governed processed record	Establish Records Processed and record-level validation status.
Exception Record Output	One row per governed exception record	Support targeted review and record-level Exception Rate.
Validation Failure Output	One row per record per failed rule	Support reason-code and multi-failure analysis.

Modeling control: Validation Failure count may be greater than Exception Record count because a single record can fail more than one rule.

9. Rule Change Control

Change	Required Control
New validation rule	Assign rule ID, document purpose, fields, logic, reason and test evidence.
Threshold change	Record the revised bound and validate boundary cases.
Code-set change	Update the governed set or mapping and retest affected logic.
Format change	Update the documented expected format and regression-test the rule.
Cross-field logic change	Document the revised relationship and test positive and negative cases.
Reason-code change	Keep reporting and exception outputs aligned with the updated classification.

10. Rule Acceptance Criteria

ID	Acceptance Condition
RA-01	Each implemented rule has a unique identifier and documented purpose.
RA-02	Applicable source fields are identified before rule execution.
RA-03	Rule logic and required parameters are documented sufficiently for independent testing.
RA-04	Failed records remain traceable to the source record and failure reason.
RA-05	Multiple rule failures do not inflate record-level Exception Records.
RA-06	Exception Rate reconciles to governed processed and exception populations.
RA-07	Power BI reporting uses the same governed exception classifications.
RA-08	No unretained healthcare thresholds, code sets or clinical rules are presented as historical client logic.

11. Portfolio Value

Why this artifact matters: This catalogue demonstrates the analytical control behind the automation: validation rules are defined, parameterized, traceable, testable and separated from record-level KPI logic. That distinction is central to a defensible high-volume data-quality solution.